QUANTITATIVE FINANCE

The Fama–French Factor Model

Why your fund's alpha might just be exposure to well-known risk factors

Muiz Olasupo · April 2026

Data: Kenneth French's Data Library

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A fund beats the market by 4%/year.

Is the manager **skilled**?

Or just buying **small**, **cheap** stocks that anyone could hold?

Fama & French showed how to tell the difference.

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01 START WITH CAPM

The Capital Asset Pricing Model says a stock's expected excess return depends on **one thing**: market beta.

$$R_i - R_f = \alpha_i + \beta_i(R_m - R_f) + \varepsilon_i$$

$R_i - R_f$ = stock excess return

$R_m - R_f$ = market excess return

β_i = sensitivity to the market

α_i = abnormal return (skill?)

If CAPM is right, $\alpha_i = 0$ for every stock. Any excess return is just **compensation for risk**.

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02 CAPM BREAKS DOWN

Two groups of stocks consistently earn returns **too high** for their beta:

Small-cap stocks

Small firms earned $\approx 2.0\%$ per year more than big firms with comparable beta (1926–2025).

Value stocks

High book-to-market firms earned $\approx 3.5\%$ per year more than growth firms (1926–2025).

These "anomalies" persist across countries and decades. CAPM cannot explain them.

Data: Kenneth French's Data Library, mba.tuck.dartmouth.edu

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03 THREE FACTORS

Fama & French (1993) add two factors to capture the premia CAPM misses:

$$R_i - R_f = \alpha_i + \beta_i^m (R_m - R_f) + \beta_i^s \, SMB + \beta_i^v \, HML + \varepsilon_i$$

Mkt–R_f Market excess return

SMB Small Minus Big — return of small stocks over large stocks

HML High Minus Low — return of value (high B/M) over growth (low B/M)

If α_i drops to zero under FF3, the fund was harvesting **factor premia** — not generating skill.

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04 HISTORICAL FACTOR RETURNS

Average annual returns (%), U.S. equity, 1926–2025

Factor	Mean	Std Dev	Sharpe	t-stat
Mkt–R_f	8.3	20.3	0.41	4.0
SMB	2.0	11.1	0.18	1.8
HML	3.5	11.8	0.30	2.9

The market premium is the largest and most reliable. The value premium (HML) has been historically strong but **weakened after 2005**. The size premium (SMB) is smaller and noisier.

Source: Kenneth French's Data Library. Mkt–R_f, SMB, HML monthly returns, annualised.

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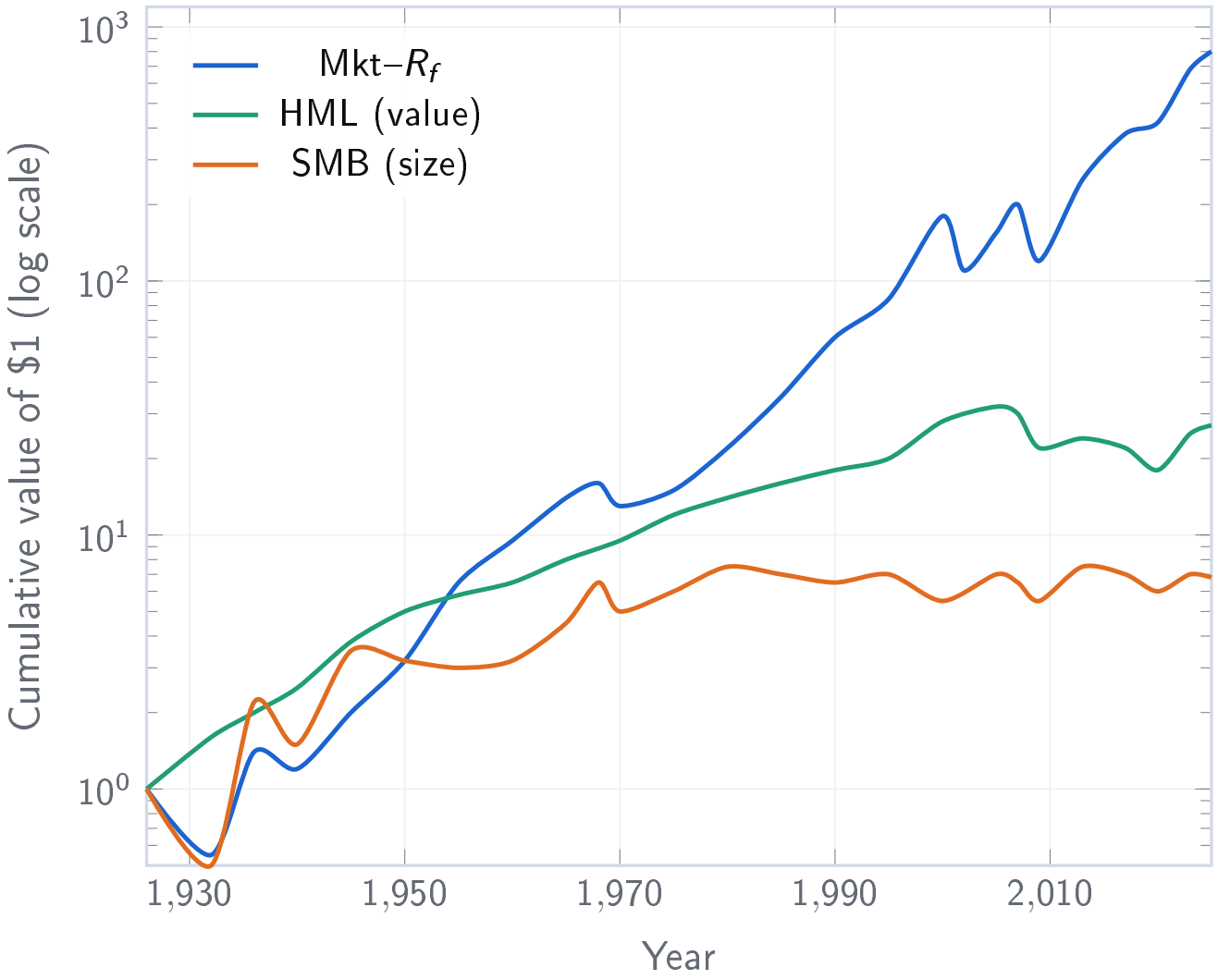
GROWTH OF $1 IN EACH FACTOR

Log scale, 1926–2025



Source: Kenneth French's Data Library. Monthly factor returns compounded.

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05 ALPHA DISAPPEARS

A fund returns 12%/year when the market returns 8%. Under CAPM that is 4% alpha. But the fund loads on small and value stocks:

	CAPM	FF3
α (annual)	**+4.0%**	**+0.3%**
β^m	1.05	0.98
β^s (SMB)	—	**0.65**
β^v (HML)	—	**0.72**
R^2	0.78	0.93

The "alpha" was just a **size premium** ($\beta^s \times$SMB) plus a **value premium** ($\beta^v \times$HML). No skill required.

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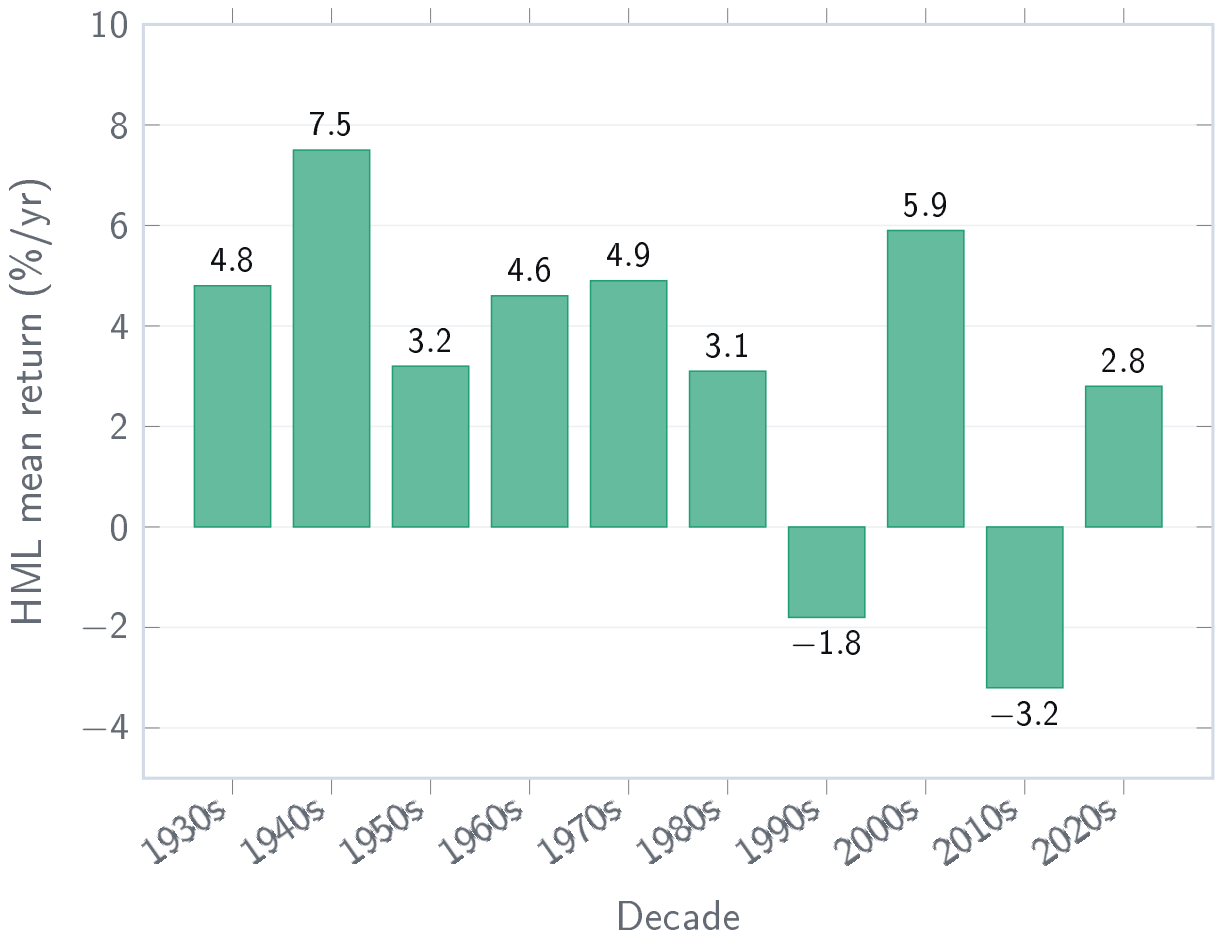
VALUE PREMIUM BY DECADE

Average annual HML return (%)



The value premium was strong for seven decades, then **turned negative** in the 1990s and 2010s. It staged a partial comeback in the 2020s — but "is value dead?" remains one of the biggest debates in finance.

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06 THE FIVE-FACTOR MODEL

Fama & French (2015) add two more factors to capture profitability and investment patterns:

$$R_i - R_f = \alpha_i + \beta_i^m (R_m - R_f) + \beta_i^s SMB + \beta_i^v HML + \beta_i^r RMW + \beta_i^c CMA + \varepsilon_i$$

RMW Robust Minus Weak — high-profit firms over low-profit firms

CMA Conservative Minus Aggressive — low-investment firms over high-investment firms

RMW $\approx$ 3.2%/yr, CMA $\approx$ 3.0%/yr (1963–2025).

Adding these two factors makes HML largely redundant — value is partly explained by profitability and investment.

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HOW SMB & HML ARE BUILT

1. **Sort all NYSE/AMEX/NASDAQ stocks**
   By market cap (size) and book-to-market (value)

2. **Form 6 portfolios** (2×3)
   Small/Big crossed with Value/Neutral/Growth

3. **Compute long–short returns**
   SMB = avg small − avg big; HML = avg value − avg growth

4. **Rebalance annually**
   End of June each year, using Dec B/M data

Source: Fama & French (1993), *J. Financial Economics*, 33(1), 3–56.

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WHY THIS MATTERS

Fund evaluation

Strip out factor exposures to isolate true manager skill. Most "alpha" vanishes under FF3/FF5.

Portfolio construction

"Smart beta" ETFs tilt toward size, value, profitability. Factor models tell you exactly what tilt you are buying.

Risk management

Decompose portfolio risk by factor. A fund concentrated in small value stocks carries size + value drawdown risk.

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THE FACTOR ZOO

Researchers have published **400+** factors. Most are spurious.

Harvey, Liu & Zhu (2016) show that the standard $t > 2$ threshold produces massive false discoveries when applied across hundreds of tested factors. They recommend $t > 3$ for new factors.

The Fama–French factors survive scrutiny because they are:

1. Persistent across 97 years of U.S. data
2. Replicated in international markets (23 countries)
3. Economically motivated by risk or mispricing theories

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Original papers

Fama, E. & French, K. (1993). Common risk factors in the returns on stocks and bonds. *J. Financial Economics*, 33(1), 3–56.

Fama, E. & French, K. (2015). A five-factor asset pricing model. *J. Financial Economics*, 116(1), 1–22.

On the factor zoo

Harvey, C., Liu, Y. & Zhu, H. (2016). . . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.

Data

Kenneth French's Data Library

`mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html`

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TAKEAWAY

Your alpha is someone else's beta.

Before calling a manager skilled,

check whether the returns are just

compensation for exposure to
size, **value**, **profitability**, and **investment**.

Fama–French gave us the tools

to decompose returns and separate
risk premia from genuine skill.

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Thank You

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Quantitative Finance

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